# Anand Bhattad

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## Research Interests

Generative Models; Neural Rendering; Inverse Graphics; Intrinsic Images; Image-based Lighting; Perceptual Organization and Physics Awareness in Large Vision Models

## Education

**University of Illinois Urbana-Champaign (UIUC)**, USA

Ph.D., Computer Science 2019 – 2024
Thesis: **Exploring Knowledge in Generative Models**
Thesis committee: **David Forsyth (Advisor)**, Alexei A. Efros, William T. Freeman, Derek Hoiem, Svetlana Lazebnik, Shenlong Wang

M.S., Computer Science 2017 – 2018
M.S., Civil Engineering 2015 – 2017

**National Institute of Technology Karnataka (NITK) Surathkal**, India

B.Tech., Civil Engineering 2011 – 2015

## Work Experience

**Toyota Technological Institute at Chicago**, Research Assistant Professor 2023 – now

**University of California Berkeley**, Visiting Scholar; Host: Alexei A. Efros 2024

**Allen Institute of Artificial Intelligence**, Research Intern with Aniruddha Kembhavi 2023

**Intel**, Research Intern with Stephan R. Richter in Vladlen Koltun's team 2021

**NVIDIA**, Research Intern with Andrew Tao and Bryan Catanzaro 2020

**Fyusion Inc**, Research Intern with Abhishek Kar and Rodrigo Ortiz Cayon 2019

**Siemens Corporate Technology**, Computer Vision & DL Intern with Jan Ernst 2017

**Michigan State University**, Research Intern with Nizar Lajnef 2014

**IIIT Hyderabad**, Research Intern with Ramacharla Pradeep Kumar 2013

## Selected Honors and Awards

**Outstanding Reviewer**, ICCV 2023

**Best Paper Nomination**, CVPR 2022

**Outstanding Emergency Reviewer**, CVPR 2021

**Excellent Teaching Assistant**, Intro Computing: Engrg & Sci 2016

## Papers Under Review

Self-selected Key Papers are Highlighted; * denotes equal contribution, and † indicates equal advising

4. **Visual Jenga: Discovering Object Dependencies through Sequential Scene Decomposition**
**Anand Bhattad**, Konpat Preechakul, Alexei A. Efros
*In Submission*

3. **LumiNet: : Latent Intrinsics Meets Diffusion Models for Indoor Scene Relighting**
Xiaoyan Xing, Konrad Groh, Sezer Karaoglu, Theo Gevers, **Anand Bhattad**
*In Submission*

2. **SpotLight: Shadow-Guided Object Relighting via Diffusion**
   Frédéric Fortier-Chouinard, Zitian Zhang, Louis-Etienne Messier, Mathieu Garon, **Anand Bhattad**, Jean-François Lalonde
   *In Submission*

1. **ScribbleLight: Single Image Indoor Relighting with Scribbles**
   Jun Myeong Choi, Annie N. Wang, Pieter Peers, **Anand Bhattad**, Roni Sengupta
   *In Submission*

## Refereed Conference Papers

15. **UrbanIR: Large-Scale Urban Scene Inverse Rendering from a Single Video**
    Zhi-Hao Lin, Bohan Liu, Yi-Ting Chen, Kuan-Sheng Chen, David A. Forsyth, Jia-Bin Huang, **Anand Bhattad**, Shenlong Wang
    *3DV 2025*

14. **ZeroComp: Zero-shot Object Compositing from Image Intrinsics via Diffusion**
    Zitian Zhang, Frédéric Fortier-Chouinard, Mathieu Garon, **Anand Bhattad**, Jean-François Lalonde
    *WACV 2025* (Oral presentation)

13. **Latent Intrinsics Emerge from Training to Relight**
    Xiao Zhang, William Gao, Seemandhar Jain, Michael Maire, David A. Forsyth, **Anand Bhattad**
    *NeurIPS 2024* (Spotlight presentation)

12. **From an Image to a Scene: Learning to Imagine the World from a Million 360° Videos**
    Matthew Wallingford, **Anand Bhattad**, Aditya Kusupati, Vivek Ramanujan, Matt Deitke, Sham Kakade, Aniruddha Kembhavi, Roozbeh Mottaghi, Wei-Chiu Ma, Ali Farhadi
    *NeurIPS 2024*

11. **Videoshop: Localized Semantic Video Editing with Noise-Extrapolated Diffusion Inversion**
    Xiang Fan, **Anand Bhattad**†, Ranjay Krishna†
    *ECCV 2024*

10. ***Shadows Don't Lie and Lines Can't Bend!* Generative Models Don't Know Projective Geometry...For Now**
    Ayush Sarkar⋆, Hanlin Mai⋆, Amitabh Mahapatra⋆, Svetlana Lazebnik, David A. Forsyth, **Anand Bhattad**
    *CVPR 2024*

9. **StyLitGAN: Image-based Relighting via Latent Control**
   **Anand Bhattad**, James Soole, David A. Forsyth
   *CVPR 2024*

8. **StyleGAN Knows Normals, Depth, Albedo, and More**
   **Anand Bhattad**, Daniel McKee, Derek Hoiem, David A. Forsyth
   *NeurIPS 2023*

7. **OBJect 3DIT: Language-guided 3D-aware Image Editing**
   Oscar Michel, **Anand Bhattad**, Eli Vanderbilt, Ranjay Krishna, Aniruddha Kembhavi, Tanmay Gupta
   *NeurIPS 2023*

6. **Improving Equivariance in State-of-the-Art Supervised Depth and Normal Predictors**
   Yuanyi Zhong, **Anand Bhattad**, Yuxiong Wang, David A. Forsyth
   *ICCV 2023*

5. **Cut-and-Paste Object Insertion by Enabling Deep Image Prior for Reshading**
   **Anand Bhattad**, David A. Forsyth
   *3DV 2022* (Spotlight presentation)

4. **DIVeR: Real-time and Accurate Neural Radiance Fields with Deterministic Integration for Volume Rendering**

Liwen Wu, Jae Yong Lee, **Anand Bhattad**, Yuxiong Wang, David A. Forsyth
*CVPR 2022* (Best Paper Nomination; Oral presentation)

3. **View Generalization for Single Image Textured 3D Models**
**Anand Bhattad**, Aysegul Dundar, Guilin Liu, Andrew Tao, Bryan Catanzaro
*CVPR 2021*

2. **Unrestricted Adversarial Perturbations via Semantic Manipulation**
**Anand Bhattad**⋆, Min-Jin Chong⋆, Kaizhao Liang, Bo Li, David A. Forsyth
*ICLR 2020*

1. **Improved Style Transfer with Calibrated Metrics**
Mao-Chuang Yeh⋆, Shuai Tang⋆, **Anand Bhattad**, Chuhang Zou, David A. Forsyth
*WACV 2020*

## Refereed Workshop Papers

5. **Intrinsic LoRA: A Generalist Approach for Discovering Knowledge in Generative Models**
Xiaodan Du, Nicholas Kolkin, Greg Shakhnarovich, **Anand Bhattad**
*CVPR-W 2024* (Oral presentation)
workshops: Generative Models for CV, AI for 3D Gen, SyntaGen, SynthData4CV, Dataset Distillation

4. **UrbanIR: Large-Scale Urban Scene Inverse Rendering from a Single Video**
Zhi-Hao Lin, Bohan Liu, Yi-Ting. Chen, David A. Forsyth, Jia-Bin Huang, **Anand Bhattad**, Shenlong. Wang
*CVPR-W 2024*
workshop: SynthData4CV

3. **MIMIC: Masked Image Modeling with Image Correspondences**
Kalyani Marathe, Mahtab Bigverdi, Nishat Khan, Tuhin Kundu, Patrick Howe, Sharan Ranjit S, **Anand Bhattad**, Aniruddha Kembhavi, Linda G. Shapiro, Ranjay Krishna
*CVPR-W* 2024
workshop: 3D with Multi-View Supervision (Archival long paper)

2. **Big but Imperceptible Adversarial Perturbations via Semantic Manipulation**
**Anand Bhattad**⋆, Min-Jin Chong⋆, Kaizhao Liang, Bo Li, David A. Forsyth
*CVPR-W 2019* (Oral presentation)
workshop: Adversarial Machine Learning in Real-World Computer Vision Systems

1. **Detecting Anomalous Faces with "No Peeking" Autoencoders**
**Anand Bhattad**, Jason Rock, David A. Forsyth
*CVPR-W 2018* (Oral presentation)
workshop: Vision with Biased and Scarce Data

## Technical Reports

4. **Generative Models: What Do They Know? Do They Know Things? Let's Find Out!**
Xiaodan Du, Nicholas Kolkin, Greg Shakhnarovich, **Anand Bhattad**
*In Submission*

3. **Blocks2World: Controlling Realistic Scenes with Editable Primitives**
Vaibhav Vavilala, Seemandhar Jain, Rahul Vasanth, **Anand Bhattad**, David A. Forsyth
Technical Report, arXiv 2023

2. **Make It So: Steering StyleGAN for Any Image Inversion and Editing**
**Anand Bhattad**, Viraj Shah, Derek Hoiem, David A. Forsyth
Technical Report, arXiv 2023

1. **SIRfyN: Single Image Relighting from your Neighbors**
David A. Forsyth, **Anand Bhattad**, Pranav Asthana, Yuanyi Zhong, Yuxiong Wang
Technical Report, arXiv 2021

## Patents

1. **Intrinsic-ControlNet: Zero-shot Object Compositing from Image Intrinsics**
   Zitian Zhang, Frédéric Fortier-Chouinard, Mathieu Garon, **Anand Bhattad**, Jean-François Lalonde
   Pending

## Mentoring

### Doctoral Students

- Jun Myeong Choi , UNC Chapel Hill, Paper under Review — 2024 – now
- William Gao, UChicago, **Paper in NeurIPS 2024** — 2024 – now
- Xiaoyan Xing, University of Amsterdam, Paper under Review — 2024 – now
- Vaibhav Vavilala, UIUC, Paper under Review — 2023 – now
- Xiao Zhang, UChicago, **Paper in NeurIPS 2024** — 2023 – now
- Matthew Wallingford, University of Washington, **Paper in NeurIPS 2024** — 2023 – now
- Xiaodan Du, TTIC, **Paper in CVPR-W 2024** & Paper under Review — 2023 – now
- Zitian Zhang, Université Laval, **Paper in WACV 2025** & Paper under Review — 2023 – now
- Xiang Fan, University of Washington, **Paper in ECCV 2024** — 2023 – now
- Ayush Sarkar, UIUC, **Paper in CVPR 2024** — 2023 – now
- Zhi-Hao Lin, UIUC, **Paper in CVPR-W 2024 & Paper in 3DV 2025** — 2023 – 2024
- Kalyani Marathe, University of Washington, **Paper in CVPR-W 2024** — 2023 – 2024

### Master's Students

- Yixin (Tracy) Zhu, UChicago, Paper under Review — 2024 – now
- Hanlin (Asher) Mai, UIUC, **Paper in CVPR 2024** — 2023 – now
- Frédéric Fortier-Chouinard, Université Laval, **Paper in WACV 2025** & Paper under Review — 2023 – now
- Seemandhar Jain, UIUC | Now: PhD at UCSD, **Paper in NeurIPS 2024** — 2023 – 2024
- James Soole, UIUC | Now: Research Engineer at MATLAB, **Paper in CVPR 2024** — 2023
- Feiran Wang, UIUC, Project on NeRF + Latent Illumination — 2023
- Pranav Asthana, UIUC | Now: PhD at UMD, Project on Relighting from Neighbors — 2021 – 2022
- Kexuan (Klaus) Zou, UIUC | Now: Software Engineer at NVIDIA, Project on 2D Meshes — 2019 – 2020

### Bachelor's Students

- Zhiyan (Alex) Wang, UChicago, Paper under Review — 2024 – now
- Joshua Ahn, UChicago, Project on Wavelets in Neural Radiance Fields — 2024
- Amitabh Mahapatra, UIUC, **Paper in CVPR 2024** & Paper under Review — 2023 – now
- Kuan-Sheng Chen, UIUC, Project on Improving Generative Models with Intrinsic Images — 2023
- Oscar Michel, AI2 | Now: PhD at NYU, **Paper in NeurIPS 2023** — 2023
- Liwen Wu, UIUC | Now: PhD at UCSD, **Paper in CVPR 2022 (Best Paper Nomination)** — 2022
- Brian Chen, UIUC | REU at CMU's RI | Next: Hive AI, Project on Lighting Correction — 2020 – 2022
- Anchu Zhu, UIUC | Next: MS in CS at USC, Project on Anomaly Detection — 2018 – 2019

### Doctoral Committees

- William Gao, "Neural Methods for 3D Mesh Editing", UChicago — 2025
- Xiao Zhang, "Representation Learning from and for Generative Models", UChicago — 2025
- Xin Yuan, "Interpretable Unsupervised Generative Learning via Factorized Architectures and Structured Bottlenecks", UChicago — 2024

## Courses Designed

**Past Meets Present: A Tale of Two Visions**, TTIC; *Course Instructor* — Spring 2024

- Developed a new course from scratch on Computer Vision, combining historical and modern approaches
- Taught a class of 20+ students from TTIC and UChicago

## Teaching Assistant

**Computer Science**, UIUC; *Graduate Teaching Assistant* 2016 – 2018

- Applied Machine Learning (CS498 AML), Fall 2018
- Data Structures (CS 225), Spring 2017
- Intro Computing: Engrg & Sci (CS 101), Spring 2016 & Fall 2016

## Invited Talks

**What Generative Image Models Understand (and Don't) about the Physical World**

- UIUC: External Speaker Series Feb 2025
- UCSD: Pixel Cafe Seminar Jan 2025

**Are Generative Image Models Physically Grounded?**

- UPenn: GRASP Seminar Dec 2024
- NYU: Guest Lecture in Saining Xie's Computer Vision Course Dec 2024

**Generative Models Inside Out**

- Midwest Computer Vision Workshop, Indiana University Sep 2024

**What Do Generative Image Models Know?**

- IIIT Hyderabad, India; Vision Seminar Jan 2024
- Exactech, Inc.; Tech Talk Oct 2023

**Exploring Knowledge in Generative Models**

- Stanford University in Jiajun Wu's group Jun 2023
- University of Tubingen, Autonomous Vision Group May 2023

**What Do Generative Models Know about Geometry and Illumination?**

- UC Berkeley: Vision Seminar Apr 2023
- NVIDIA Research Apr 2023
- MIT: Vision and Graphics Seminar Apr 2023
- CMU: VASC Seminar Mar 2023
- UW: RAIVN Vision Seminar Mar 2023

**Towards a Productive and Fun PhD Experience**

- UIUC: Computer Vision Workshop Apr 2023

**Learning about Light Without Labeled Data**

- UMD: Vision Seminar Mar 2023
- UCSD: Pixel Cafe Seminar Feb 2023
- TTIC: Research Talk Feb 2023

**Drag-and-Drop Rendering: Towards In the Wild Image Editing**

- CMU: Misc-Read Vision Reading Group Aug 2021

## Services & Leadership

### Reviewing

- **Area Chair**, ICCV 2025, CVPR 2025, WACV 2025 2024 – now
- **Reviewer**, CVPR, NeurIPS, ECCV, ICCV and PAMI 2018 – now

### Community Building Workshops

- **Lead Organizer**, "How to Stand Out in the Crowd?" workshop at CVPR Jun 2025
- **Lead Organizer**, "CV 20/20: A Retrospective Vision" workshop at CVPR Jun 2024
- **Lead Organizer**, "Scholars & Big Models: How Can Academics Adapt?" workshop at CVPR Jun 2023

## Technical Workshops

- **Co-Organizer**, "Ind3D: Enforcing geometric, physical, topological, and functional inductive bias in 3D generation" workshop at CVPR — Jun 2025
- **Lead Organizer**, "Knowledge in Generative Models" workshop at ECCV — Sep 2024
- **Co-lead Organizer**, "Multimodal Artificial Intelligence " workshop at TTIC — Aug 2024
- **Lead Organizer**, Computer Vision workshop at Allerton, UIUC — Apr 2023
- **Committee**, "Adversarial ML in Real-World Computer Vision Systems", workshop at CVPR — Jun 2019
- **Committee**, "Security and Privacy in Machine Learning" workshop at ICML — Jun 2019

## Seminars and Reading Groups

- **Organizer**, UIUC Vision Seminar, a new weekly seminars inviting speakers outside UIUC — 2021 – 2023
- **Organizer**, Vision Lunch (computer vision reading group) — 2021 – 2023
- **Organizer**, UIUC Summer Vision Coffee — 2018

**Co-Founder**, SchoolEngg — 2013 – 2015

- Forum for high-school students; providing firsthand knowledge of all engineering disciplines
- Renamed as *PrepLift*, an education counselling start-up in India

**Co-Founder**, American Society of Civil Engineers (ASCE) NITK Chapter — 2013 – 2015

- Co-Founded the *first* ASCE student chapter of India
- Served as treasurer for one year and as a senior advisor for the second year
- Coordinated with board members, faculty, and outside organizations to facilitate 10+ events